

AIGR Focused Review — rad3 (SCHPO, Q02099) nucleolus GO:0005730

Gene: rad3 (SPBC216.05) · **Organism:** *Schizosaccharomyces pombe* 972 / ATCC 24843 (NCBITaxon:284812) · **UniProt:** Q02099 **Focus:** existing GO annotation decision — `existing_annotations[36]` of `genes/SCHPO/rad3/rad3-ai-review.yaml` **Hypothesis slug:** `existing-go-0005730-mark-as-over-annotated` **Annotation under review:** nucleolus (GO:0005730), IDA, qualifier `is_active_in`, reference

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Seed review action: MARK_AS_OVER_ANNOTATED

Summary

The seed hypothesis is **SUPPORTED**, and on closer inspection the annotation is even weaker than "over-annotated" implies — it is a candidate for outright **REMOVAL** or **reference correction**. The rad3→nucleolus (GO:0005730) association rests on a **single, uncorroborated IDA annotation** whose only cited primary source,

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(Bailis et al. 2008, *Mol. Cell. Biol.*, PMC2258774), does not localize Rad3 to the nucleolus — or anywhere. Full-text mining of that paper (~97,000 characters covering Abstract, Methods, Results, Discussion and References) recovered **zero** occurrences of "nucleol-", "rDNA", "ribosom-", or "GFP", and the only Rad3 mentions (4 total) are genetic/pathway references (Rad3–Cds1 checkpoint, Δ rad3 mutant, Rad3 required for Cds1 activation, and one citation). The proteins actually imaged in that study are Rad22-YFP, Rhp51, Mcm4, Rad52, Crb2 and phospho-H2A. **Rad3 itself was never subjected to a localization assay.** This strongly indicates the annotation is a **mis-attributed reference or a curation error**, not a genuine minor or context-specific pool.

Independent lines of evidence corroborate the over-annotation call. QuickGO returns exactly **one** annotation linking Q02099 to GO:0005730 (IDA/ECO:0000314, assigned by PomBase, qualifier `is_active_in`); no second, independent primary source exists. The genome-wide YFP

ORFeome

localization

screen

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P 16823372
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places Rad3 in the **cytosol/diffuse** signal, not the nucleolus. The intensively studied human ortholog ATR (Q13535) is **not** curated to the nucleolus; its curated compartments are nucleoplasm, chromosome, PML body, nuclear envelope and sites of DNA damage. Where ATR-family kinases do engage the nucleolus in mammals, the activity is demonstrably **stress-induced and context-specific** (TopBP1 overexpression, RNA Pol I inhibition), not a constitutive core compartment.

Rad3's established, multiply corroborated sites of action are the **ATR–ATRIP (Rad3–Rad26) complex, chromatin/stalled replication forks, and telomeres** — each supported by multiple primary references with IDA evidence. The is_active_in qualifier on the nucleolus term is particularly problematic because it asserts *functional activity* in a compartment where neither the cited paper nor any other primary source demonstrates Rad3 presence, let alone function. **Recommended lead: REMOVE or correct the reference; MARK_AS_OVER_ANNOTATED is fully supported as a conservative floor**, pending a curator's inspection of the journal PDF and supplemental figures to exclude a reference typo or overlooked panel.

Executive Judgment

Verdict: SUPPORTED — and the underlying evidence is weaker than the seed states. The over-annotation call is justified; the evidence in fact points toward outright **removal / reference correction**, subject to a curator's check of the published PDF and supplemental figures.

The decisive result is the full-text mining of the cited reference (Finding 2). Because the annotation carries evidence code **IDA/ECO:0000314** ("inferred from direct assay"), it requires a demonstrated experimental localization of Rad3 in the cited paper. No such assay exists in

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The annotation therefore **fails its own evidence-code requirement**. The strongest residual caveat is that retrieval was of the PMC HTML article, so a small chance remains that a separately hosted supplemental figure contains Rad3 data — although the main text and figure legends never mention the nucleolus and give no indication Rad3 was imaged.

Key Findings

Finding 1 — The nucleolus annotation is single-source and non-core; the over-annotation call is justified

Rad3's curated cellular-component (CC) annotations span several compartments, but they are not of equal evidential weight. Assembling the PomBase curated set for SPBC216.05 / Q02099:

Compartment (GO CC)	Evidence	Supporting references	Corroboration
ATR–ATRIP complex (GO:1990635)	IDA	P 10559981	Core complex — the Rad3–Rad26 heterodimer
chromatin (GO:0000785)	IDA	P 17531813 , P 21945095	Multiple references
telomeric repeat-containing region	IDA	P 12196391 , P 20140190	Multiple references
nuclear chromosome	IDA	P 8843195	Consistent with chromatin role
cytosol	HDA	P 16823372	Genome-wide ORFeome screen
nucleolus (GO:0005730)	IDA	P 18180284 (only)	None — single source

The nucleolus term is the **only** CC annotation resting on a single IDA, and its source paper (Bailis et al. 2008) is primarily about the MCM complex and replication-fork protection; its abstract does not mention the nucleolus. The genome-wide YFP ORFeome study ([P 16823372](#)) that systematically localized *S. pombe* proteins assigned Rad3 to a **cytosolic/diffuse** signal — an orthogonal, high-throughput dataset that fails to reproduce a nucleolar pool. The deeply

studied human ortholog **ATR (Q13535)** carries curated CC terms for nucleoplasm, chromosome, PML body, nuclear envelope and sites of DNA damage, but **no nucleolus term**, despite decades of study.

Where mammalian ATR/ATM signaling has been documented at the nucleolus, it is explicitly **inducible and stress-dependent**: TopBP1 overexpression triggers ATR-dependent shut-down of rRNA transcription and nucleolar segregation (P 25916852),

and Pol I transcription inhibition (CX-5461) activates non-canonical ATM/ATR signaling within nucleoli (P 27391441).

These are context-specific responses, not evidence of a constitutive nucleolar compartment for the checkpoint kinase. Taken together, the nucleolus annotation is single-source, non-core, and inconsistent with orthogonal localization data and ortholog curation — **MARK_AS_OVER_ANNOTATED is well supported.**

Finding 2 — The cited reference

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contains no Rad3 localization and never mentions the nucleolus; the annotation is likely mis-attributed

This is the decisive finding. Full text of

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(Bailis et al. 2008, PMC2258774; ~97,000 characters including Abstract, Methods, Results, Discussion and References) was mined directly:

- nucleol → 0 occurrences
- rDNA → 0 occurrences
- ribosom → 0 occurrences
- GFP → 0 occurrences
- Rad3 → 4 occurrences, all as genetic/pathway references (the Rad3–Cds1 checkpoint; the Δrad3 mutant; Rad3 required for Cds1 activation; one bibliographic citation). None occurs within 60 characters of "foci", "focus", "localiz", "nucle", "stain", "YFP" or "GFP".

The proteins actually imaged in the paper are **Rad22-YFP, Rhp51, Mcm4, Rad52, Crb2 and phospho-H2A**. Rad3 itself is never subjected to a localization assay. The PomBase annotation detail (id 433558) records IDA / ECO:0000314, low-throughput, assigned_by PomBase on 2015-11-25, with no extension and only the `is_active_in` qualifier. Because IDA/ECO:0000314 demands a demonstrated experimental localization in the cited paper, and no such assay for Rad3 exists in

P 18180284

the annotation fails its own evidence-code requirement. The two most parsimonious explanations are (1) a mis-attributed/typo'd reference (the wrong PMID was entered), or (2) a curation slip in which text about a different imaged protein was mapped to Rad3. Either way, the citation does not support the term.

Finding 3 — QuickGO confirms a single, uncorroborated `is_active_in` nucleolus annotation

An independent QuickGO annotation search (geneProductId=Q02099, goId=GO:0005730) returns **numberOfHits** = **1**: IDA / ECO:0000314, reference

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assignedBy PomBase, qualifier `is_active_in`. There is no second or independent annotation from any other database or reference. Targeted PubMed searches for an alternative primary paper reporting *S. pombe* Rad3 nucleolar localization returned none. The `is_active_in` qualifier is especially problematic: it asserts that Rad3 is **functionally active** in the nucleolus, a claim demanding functional evidence there. The cited paper provides neither localization nor function, so an `is_active_in nucleolus` assertion is unsupported on its own terms.

Mechanistic Model / Interpretation

The entity under evaluation is **Rad3**, the *S. pombe* ATR-family PIKK Ser/Thr checkpoint kinase (InterPro domains: PI3/4-kinase catalytic, FAT, FATC, HEAT-ATR, UME). It functions as the Rad3–Rad26 (ATR–ATRIP) heterodimer — the sensor apex of the DNA replication/damage checkpoint — sensing stalled replication forks and DNA damage and activating downstream effector kinases Cds1 and Chk1. The GO term in question, nucleolus (GO:0005730), asserts a cellular location (and, via `is_active_in`, a functional site) that must be judged against where Rad3 is actually demonstrated to reside and act.

Rad3 established sites of action (multiply corroborated)

Rad3–Rad26 (ATR–ATRIP) complex	— https://pubmed.ncbi.nlm.nih.gov/10559981/
chromatin / stalled forks	— https://pubmed.ncbi.nlm.nih.gov/17531813/
telomeric repeat region	— https://pubmed.ncbi.nlm.nih.gov/12196391/
nuclear chromosome	— https://pubmed.ncbi.nlm.nih.gov/8843195/

| activates



Cds1 / Chk1 effector kinases



Downstream: Clp1/Flp1 released FROM the nucleolus on stress

(<https://pubmed.ncbi.nlm.nih.gov/18385517/> title="Visit P

NOT evidence that Rad3 itself resides in the nucleolus.

Nucleolus (GO:0005730) annotation

Single IDA · https://pubmed.ncbi.nlm.nih.gov/18180284/
→ cited paper: 0 mentions of nucleolus/rDNA/ribosom/GFP
→ Rad3 never imaged; proteins imaged = Rad22, Rhp51, Mcm4...
→ orthogonal ORFeome screen: Rad3 = cytosol/diffuse
→ human ATR ortholog: NOT curated to nucleolus

A crucial conceptual distinction is that **downstream substrates of the Rad3 pathway do relocate at the nucleolus**. The Cdc14-family phosphatase Clp1/Flp1 is nucleolar-sequestered and is released into the nucleoplasm on replication stress in a Cds1/checkpoint-dependent manner

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[P 18385517](https://pubmed.ncbi.nlm.nih.gov/18385517/)

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[P 22918952](https://pubmed.ncbi.nlm.nih.gov/22918952/)

This is genuine and interesting nucleolar biology — but it concerns the **substrate (Clp1/Flp1)**, not Rad3 itself. It would be an error to infer Rad3 nucleolar residence from the nucleolar release of one of its pathway's downstream targets. This substrate-versus-kinase conflation is a plausible route by which a nucleolar annotation could have entered Rad3's record. The correct GO representation of Rad3's function localizes it to chromatin, forks and telomeres — the sites where the kinase is directly demonstrated to bind and act.

Evidence Base / Evidence Matrix

Citation	Evidence type	Stance	Claim tested	Key finding	Context	Confid limitat
PMID:18180284	Localization (IDA) — annotation source	Refutes / not supported	rad3 in nucleolus	Full-text mining: 0 mentions of "nucleolus"/"rDNA"; Rad3 never imaged (only Rad22, Rhp51, Mcm4, Rad52, Crb2, γH2A localized). Cited paper does not support any Rad3 localization	<i>S. pombe</i> , MCB 2008	Mediu High: comple main t mined chars, PMC22 residu from separa hosted supple
PMID:16823372	Genome-wide localization (HDA)	Refutes core nucleolar call	Steady-state rad3 location	ORFeome YFP screen assigns rad3 to cytosol/diffuse , not nucleolus	<i>S. pombe</i> , systematic	Mediu High: system but low resolut tag/ expres caveat
PMID:17531813	Interaction/localization (IDA)	Supports core = chromatin	rad3 site of action	"Cdc18 persists in a chromatin-bound complex including the checkpoint kinases Rad3 and Rad26"	<i>S. pombe</i>	High
PMID:12196391	Localization (IDA)	Supports core = telomere	rad3 site of action	Checkpoint sensors incl. rad3 bind telomeres	<i>S. pombe</i>	High
PMID:20140190	Recruitment/localization (IDA)	Supports core = chromatin/telomere	rad3 site of action	Kinase-independent Rad3–Rad26 recruitment role	<i>S. pombe</i>	High
PMID:10559981					<i>S. pombe</i>	High

Citation	Evidence type	Stance	Claim tested	Key finding	Context	Confidence/limitations
	Complex (IDA)	Supports core assembly	Rad3–Rad26 complex	ATR–ATRIP (Rad3–Rad26) complex responds to DNA damage		
UniProt Q13535 (human ATR)	Curated database	Supports over-annotation	Is nucleolus a curated ATR compartment?	ATR CC = nucleoplasm, chromosome, PML body, nuclear envelope, DNA-damage sites — no nucleolus	Human, curated	Medium: ortholog inference, absence weaker, positive evidence
PMID:25916852	Mechanism (mammalian)	Qualifies (context-specific)	ATR at nucleolus	"TopBP1-induced nucleolar segregation is coupled to shut-down of ribosomal RNA transcription in an ATR-dependent manner"	Human cells	Medium: but inconsistent
PMID:27391441	Mechanism (mammalian)	Qualifies (context-specific)	ATR/ATM in nucleolus	Pol I inhibition (CX-5461) activates ATM/ATR signaling within nucleoli	Human cells	Medium: stress-induced
PMID:18385517	Mechanism/localization	Competing/alternative	Is nucleolar biology about Rad3 or its substrate?	"Active Cds1p checkpoint kinase is required to release Flp1p into the nucleus" — a substrate event	<i>S. pombe</i>	High: Cds1p nucleolar is downregulated substrate
PMID:22918952	Mechanism/localization	Competing/alternative	Same as above	Rad3-effector kinases Cds1/Chk1 phosphorylate Clp1 to drive	<i>S. pombe</i>	High: reinforces substrate

Citation	Evidence type	Stance	Claim tested	Key finding	Context	Confidence/limitations
				nucleoplasmic accumulation		relocalization interpretation

(Computed provenance from the iterations was saved as `evidence_matrix.csv`, `rad3_CC_provenance.csv`, and `GO_decision_table.csv`.)

GO Curation Implications

Lead (requires curator verification): remove or correct the reference for GO:0005730 (nucleolus, CC) on rad3.

- **Term class:** GO:0005730 is a cellular-component term. The question is whether the nucleolus is even a supported localization of Rad3, and whether it is *core*. The weight of evidence says the cited support is **absent** and the nucleolus is **not core** — chromatin, forks, telomeres and the ATR–ATRIP complex are the core CCs.
- **Primary recommendation:** REMOVE (IDA/ECO:0000314 is not satisfied by

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— the paper neither images Rad3 nor mentions the nucleolus). At minimum, the reference is wrong and must be corrected.

- **Fallback (if an inaccessible supplemental figure does show Rad3 in the nucleolus):** MARK_AS_OVER_ANNOTATED / KEEP_AS_NON_CORE with a context/stress qualifier — still not a core compartment.
- **is_active_in qualifier:** This is the strongest form of location assertion (functional activity in the compartment) and is the least defensible here — it should be dropped along with the term.
- **What the evidence *does* support (unchanged):** chromatin (GO:0000785), telomeric repeat-containing region, nuclear chromosome, and the ATR–ATRIP kinase complex (GO:1990635) remain well corroborated.
- **Avoided fallback:** No "protein binding" fallback is invoked; the recommendation is specific.
- **Bottom line:** The seed action MARK_AS_OVER_ANNOTATED is supported and, if anything, conservative.

Mechanistic Scope

The immediate molecular function of Rad3 is a **PIKK Ser/Thr protein kinase** acting as the sensor apex of the DNA replication/damage checkpoint, functioning as the Rad3–Rad26 (ATR–ATRIP) complex. Its direct action occurs on **chromatin at stalled replication forks and at telomeres**, where it phosphorylates downstream effectors (Cds1, Chk1). Direct gene-product activity (the kinase acting on chromatin/forks/telomeres) is not in dispute. What is in dispute is whether Rad3 *resides and is active in the nucleolus* — and the available primary evidence does not place it there. The nucleolar phenomena in this pathway that are real (Clp1/Flp1 release from the nucleolus; mammalian ATR-dependent rRNA shut-down) are either **downstream substrate relocalizations** or **stress-induced context-specific responses of orthologs**, not constitutive Rad3 nucleolar residence. Any nucleolar/rDNA role (as inferred from mammalian ATR) would be a **downstream, stress-inducible** activity, not the primary function.

Conflicts and Alternatives

1. **Substrate–kinase conflation (most likely).** The Rad3 pathway has a bona fide nucleolar node — Clp1/Flp1 sequestration and checkpoint-dependent release

(
P 18385517,
P 22918952).
 A curation step that linked "nucleolar checkpoint biology" to Rad3 rather than to its substrate would produce exactly this annotation. This alternative reinforces removal.

2. **Reference mis-attribution / typo (strongest alternative, now evidenced).** Full-text mining of

P 18180284
 found no Rad3 localization experiment and no mention of the nucleolus. The most parsimonious explanation is a mis-attributed reference or curation slip — the annotation may have been intended for a different paper, or the curator conflated Rad3 with another imaged protein. If confirmed, this favours **REMOVE**.

3. Genuine but context-specific nucleolar biology of ATR-family kinases

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P 25916852
P 27391441
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ATR is recruited to rDNA/nucleoli during nucleolar stress and Pol I inhibition, so the term is *not biologically absurd* — it argues for "non-core/context-specific" rather than "impossible," if any signal survives verification.

4. **Systematic vs. targeted disagreement.** The ORFeome screen (cytosol) and the single IDA (nucleolus) disagree; both are low-to-medium resolution for a diffuse nuclear kinase. Neither establishes the nucleolus as a preferred compartment.
5. **No paralog confusion in *S. pombe*.** Rad3 is the single-copy PIKK-ATR; Tel1 is the ATM ortholog, separately annotated. Cross-organism carry-over is also unlikely, since human ATR is itself not curated to nucleolus.

Limitations and Knowledge Gaps

1. Exact content of

P 18180284
for Rad3. Checked: abstract and complete main text/legends/references via PMC HTML (PMC2258774) — "nucleolus" and "rDNA" absent; Rad3 never imaged. Remaining unknown: separately hosted **supplemental figures** could not be programmatically retrieved. This matters because it decides REMOVE vs. non-core. Resolve by opening the journal PDF + supplemental files and confirming no Rad3-nucleolus panel exists.

2. **Independent *S. pombe* evidence for nucleolar Rad3.** Checked PomBase CC set and ORFeome — none corroborate the nucleolus. A second independent localization (endogenous tag, high resolution, or under replication stress) would resolve whether a minor nucleolar pool is reproducible.
3. **Functional relevance at rDNA in fission yeast.** No *S. pombe* assay ties Rad3 catalytic function to rDNA/nucleolar processes. A Rad3-dependent phenotype at rDNA (fork stalling, rDNA stability, Pol I regulation) would upgrade the term from CC-only to functionally meaningful.

4. **Negative-evidence weight of ortholog curation.** ATR lacking a nucleolus CC term is suggestive but is absence-of-evidence, not evidence-of-absence.

Discriminating Tests

- **Re-read**

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localization figures/legends (main + supplemental) to confirm whether any Rad3 image panel exists and, if so, its marker, condition and quantitation.

- **Endogenous high-resolution imaging** of Rad3-GFP/mNeonGreen (native locus) with a nucleolar marker (Gar2/Nop1/fibrillarin) and rDNA FISH, $\pm$ replication stress (HU) / rDNA fork-block, to test for a reproducible minor nucleolar pool vs. general nucleoplasmic signal.
- **rDNA-directed ChIP / CUT&RUN** of Rad3–Rad26 to test physical association with rDNA repeats.
- **Curation provenance audit:** retrieve PomBase annotation id 433558 history and original curator note to verify the intended reference and figure.
- **Functional test:** rad3 Δ (or kinase-dead) effect on rDNA stability / Pol I regulation to determine whether any nucleolar presence is functional.

Proposed Follow-up Actions / Curation Leads (require curator verification)

- **Primary action lead:** REMOVE GO:0005730 (nucleolus, IDA,

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— or, at minimum, flag the reference as not supporting the annotation and request re-curation. The seed's **MARK_AS_OVER_ANNOTATED** is fully supported as a floor.

- **Decisive verifiable evidence:** Full text of PMC2258774 mined — "nucleolus"/"nucleolar" = 0 occurrences; "rDNA"/"ribosom-" = 0; "Rad3" = 4 occurrences, all genetic/pathway, none localization; imaged proteins = Rad22-YFP, Rhp51, Mcm4, Rad52, Crb2, phospho-H2A (Rad3

absent). *Curator: reproduce by searching the article PDF for "nucleol" and for any Rad3 image panel.*

- **Candidate supporting references / snippets to verify:**

- [P 16823372](#)

(ORFeome) — systematic localization assigns rad3 to cytosol, not nucleolus.

- [P 17531813](#)

— "*Cdc18 persists in a chromatin-bound complex including the checkpoint kinases Rad3 and Rad26.*" (core = chromatin).

- [P 18385517](#)

— "*Active Cds1p checkpoint kinase is required to release Flp1p into the nucleus*" (nucleolar biology here is a substrate event, not Rad3 residence).

- UniProt Q13535 — human ATR curated CC lacks a nucleolus term.

- [P 25916852](#)

- [P 27391441](#)

— ATR-family nucleolar activity is stress/context-specific (rationale for "non-core," not "impossible").

- **Suggested curator questions:** (1) Does

- [P 18180284](#)

actually contain any Rad3 localization panel (main or supplemental)? (2) If not, was the intended reference a different paper, and which? (3) Should PomBase annotation id 433558 be retracted or re-referenced?

- **Suggested experiment (only if a genuine signal must be resolved):** Endogenous Rad3-GFP + nucleolar marker (Gar2/fibrillarin) colocalization ± HU (see Discriminating Tests).

Bottom line

The seed action **MARK_AS_OVER_ANNOTATED** is **supported**, and full-text mining of the cited reference (PMC2258774) shows the evidence is weaker still: the paper never mentions the nucleolus and never localizes Rad3, so the annotation is most likely a **mis-attributed**

reference / curation error and a candidate for REMOVE. Rad3's core compartments are the ATR-ATRIP complex, chromatin/stalled forks and telomeres. A curator should confirm against the journal PDF + supplemental figures before finalizing REMOVE vs. non-core.

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